WSe2: structure-driven seven-term pilot

STATUS: Not accepted as a replacement

17 actual relaxed structures, 1.20286--5.09 degrees. Heff is unchanged.

Hamiltonian: quadratic kinetic + first-shell even/odd potential + odd third-shell potential

+ first/second tunneling harmonics + one real linear momentum-tunneling term.

The old even second-shell V2 is replaced by a layer-odd sine harmonic at 2g.

ONE shared mass; 15 global material coefficients (old empirical curve: 23).

$p(d) = p_0 + p_1(d - 6.65\,\mathrm{A}) + p_2(d - 6.65\,\mathrm{A})^2$

$K_-(q, \bar{d}) = (1 - e^{-q\bar{d}})/q$

FE and PE responses jointly generate the odd first AND odd third shell.

One distance-dependent local tunneling generates BOTH w1 and w2.

No polynomial in angle; each new angle requires its own relaxed structure.

Shared mass = 0.408097 me

Full-path mean angle RMS3: old 0.712 meV; physical 1.635 meV

Small-angle HS worst principal overlap squared = 0.89121

Acceptance uses energy caps, dominant C3 ordering, and small-angle HS principal overlap.

Passing these gates would not establish that every isolated band has high overlap.

This pilot does not locate new Chern transitions or claim endpoint extrapolation robustness.

Hamiltonian and structural response law

$$H_K^{bb/tt} = -aD_{b/t}^2 + V_{b/t}, \qquad H_K^{bt} = T_0 + \delta T, \quad H_K^{tb} = (H_K^{bt})^\dagger$$

$$V_{b/t} = 2\sum_{j=1}^{3}[V_c \cos(\mathbf{g}_j \cdot \mathbf{r}) \mp V_s \sin(\mathbf{g}_j \cdot \mathbf{r})] \mp 2U_3 \sum_{j=1}^{3} \sin(2\mathbf{g}_j \cdot \mathbf{r})$$

$$T_0 = -w_1 F_0 - w_2 F_2, \quad \delta T = -\lambda e^{i\theta/2}[D_{b,+} F_+ + F_- D_{t,-}]$$

$$p(d) = p_0 + p_1(d - d_0) + p_2(d - d_0)^2, \quad d_0 = 6.65\,\mathrm{A}$$

$$V_c = \sum_n \varepsilon_n E_{1n}[u, d] + \zeta S_1^+[u]$$

$$V_s = \sum_n f_n F_{1n}[u, d] + e_{11} P_1[u, d] + \zeta S_1^-[u]$$

$$U_3 = \sum_n f_n F_{3n}[u, d] + e_{11} P_3[u, d] + \zeta S_3^-[u]$$

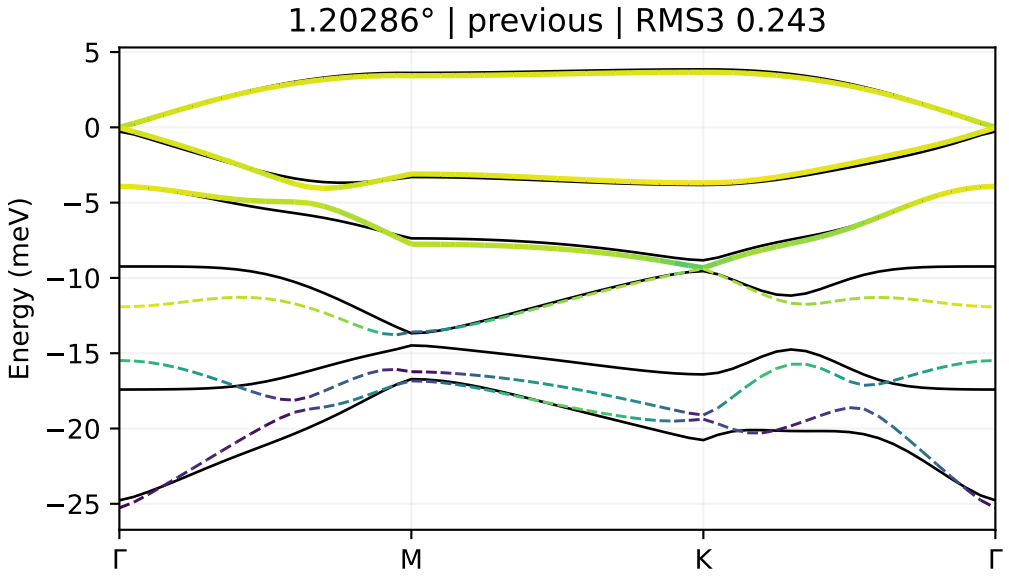
$$w_1 = \sum_n t_n T_{1n}[u, d], \quad w_2 = \sum_n t_n T_{2n}[u, d], \quad \lambda = \sum_n c_n L_n[u, d]$$

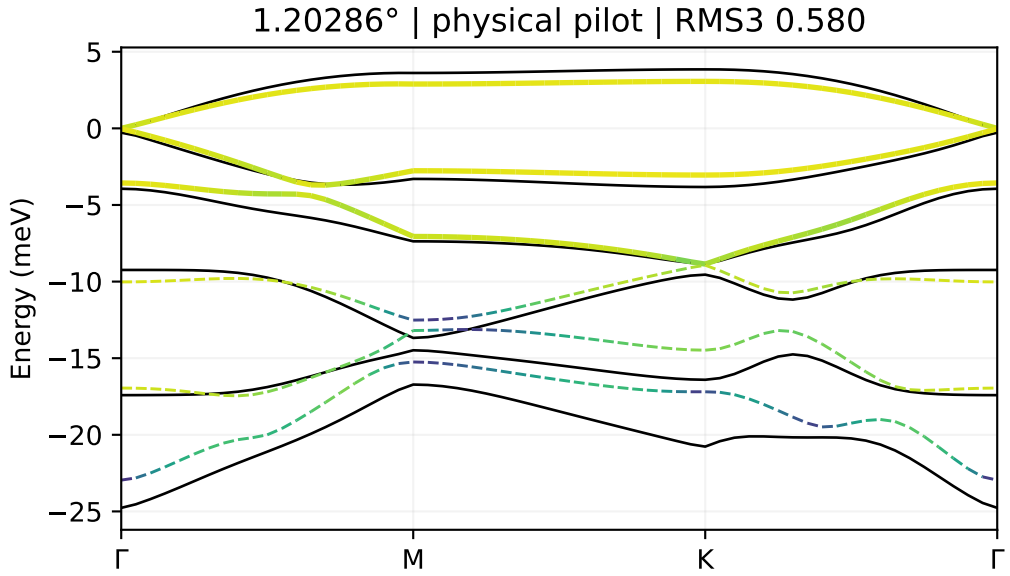
Real-space signs above use plane waves exp[i(k-Q).r]. The third shell has radius 2|g|.

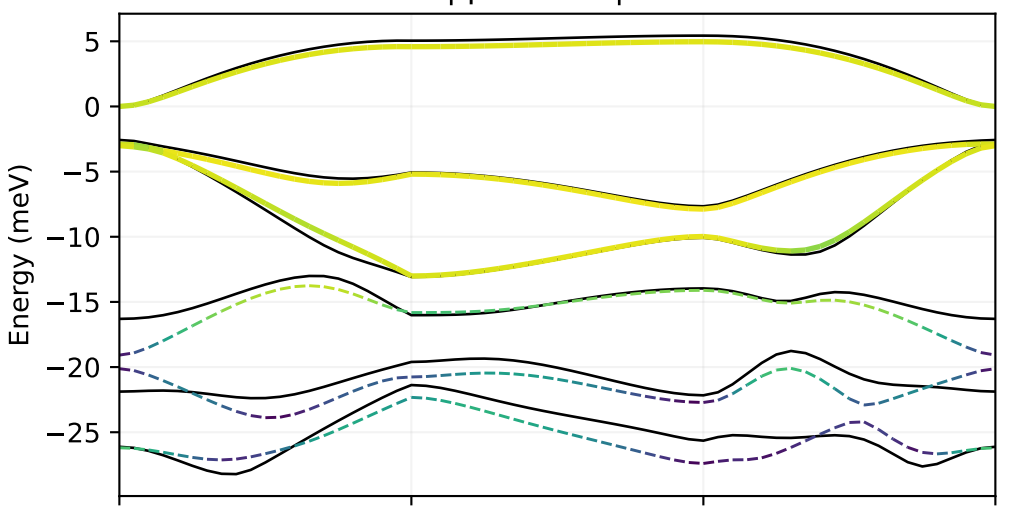
E,F,P,S,T,L are computed from each actual relaxed structure; material coefficients are shared.

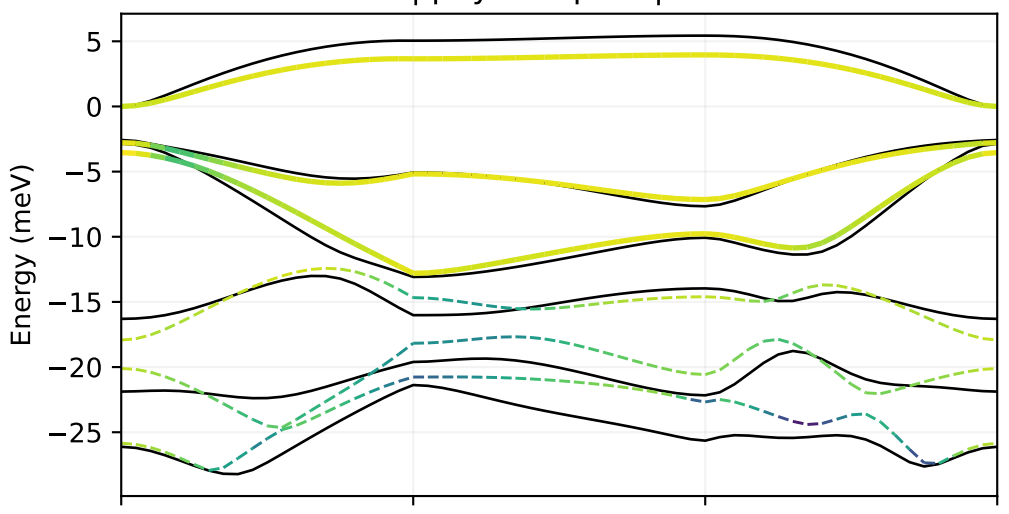
Approximations: leading registry charge ansatz, mean-separation Poisson kernel, selected harmonic truncation.

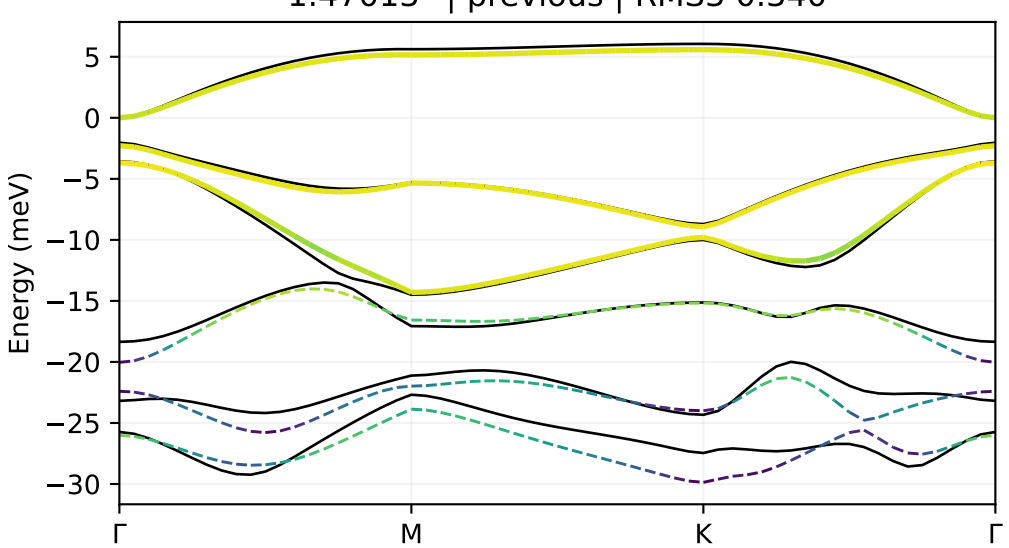
Black: unchanged Heff | colored: model top6 | solid: top3, dashed: 4--6

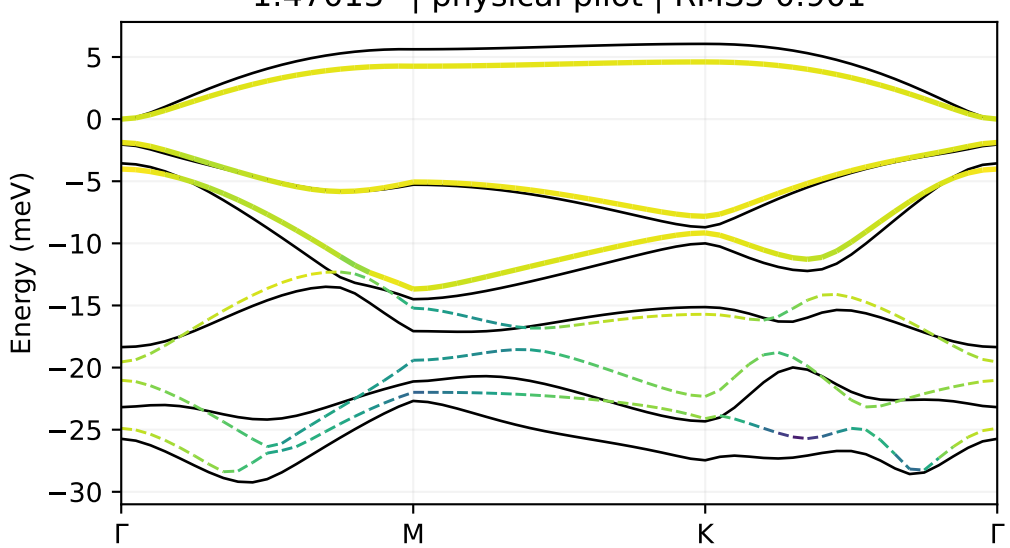


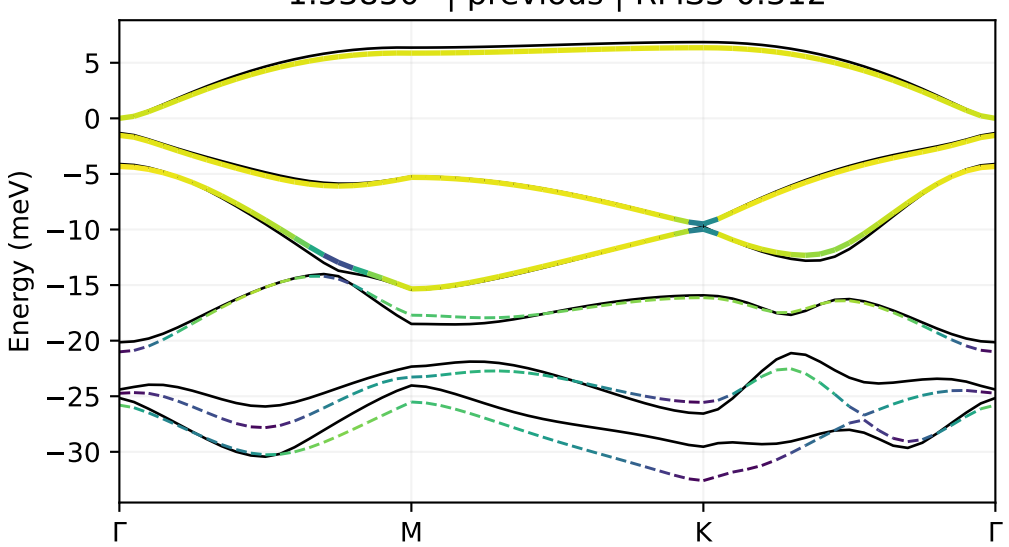


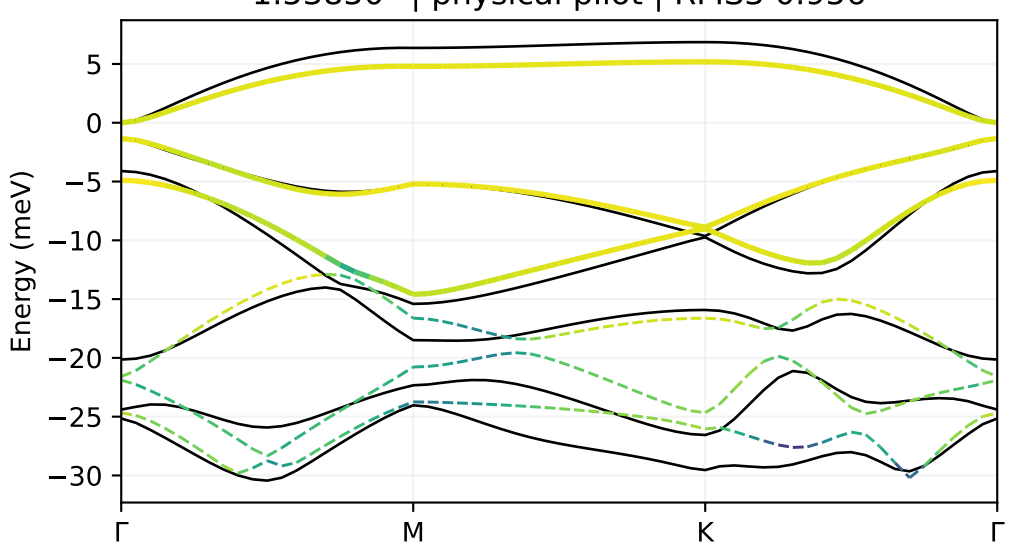


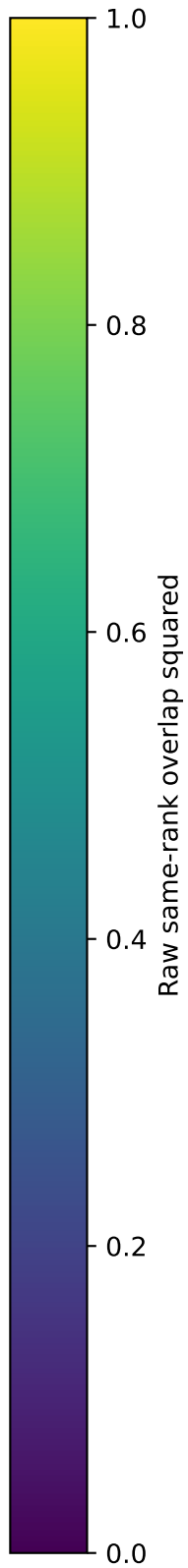


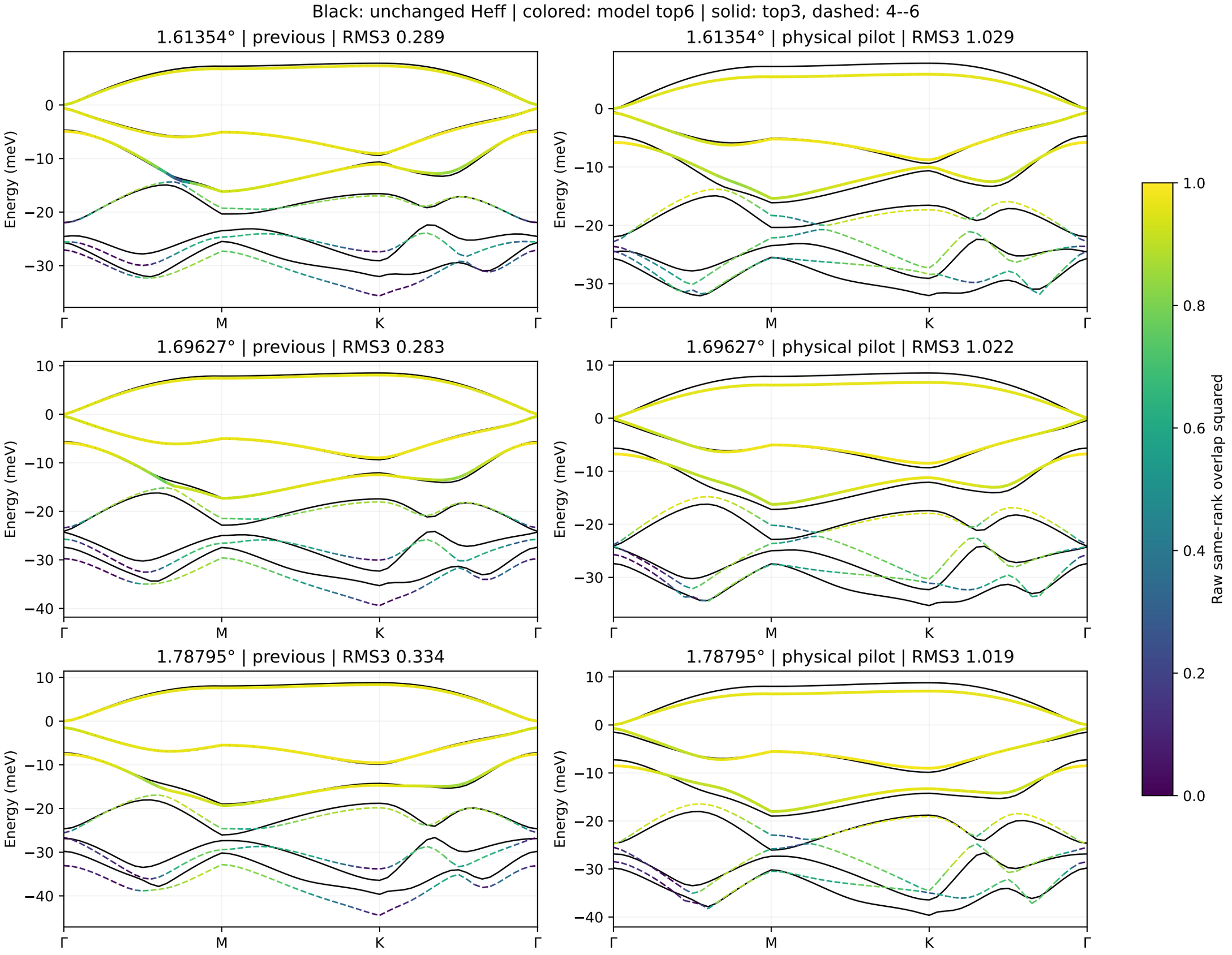




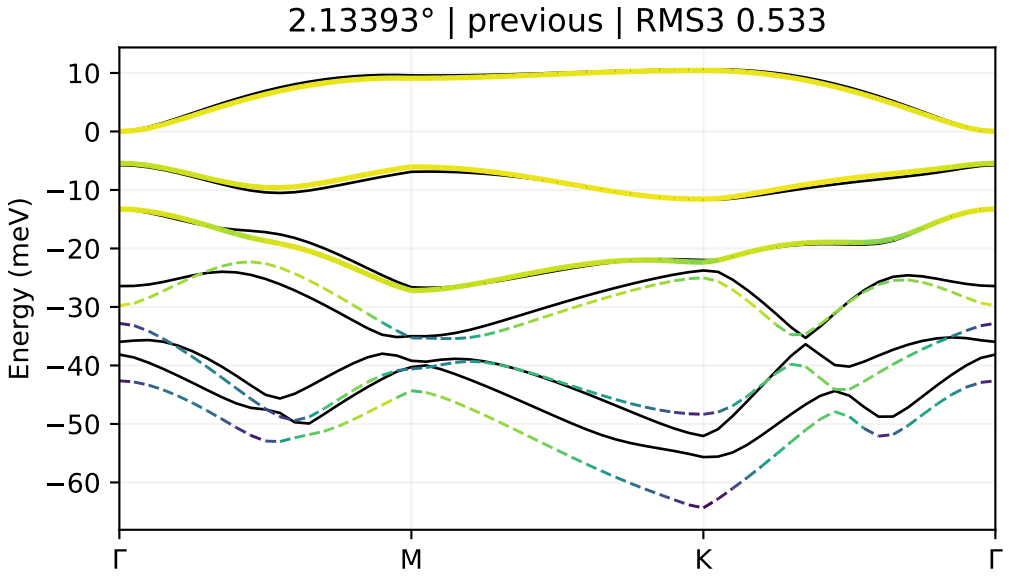


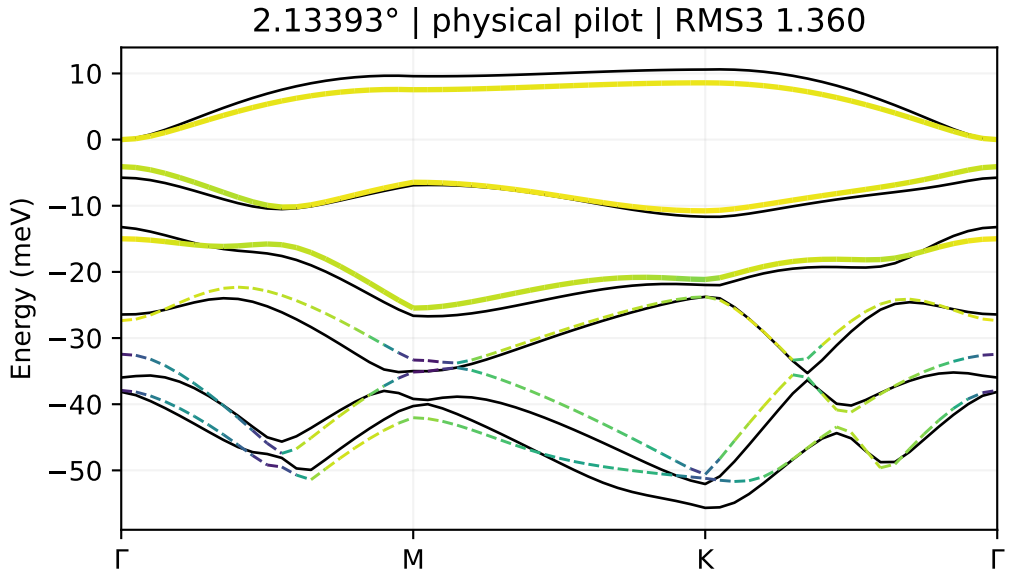


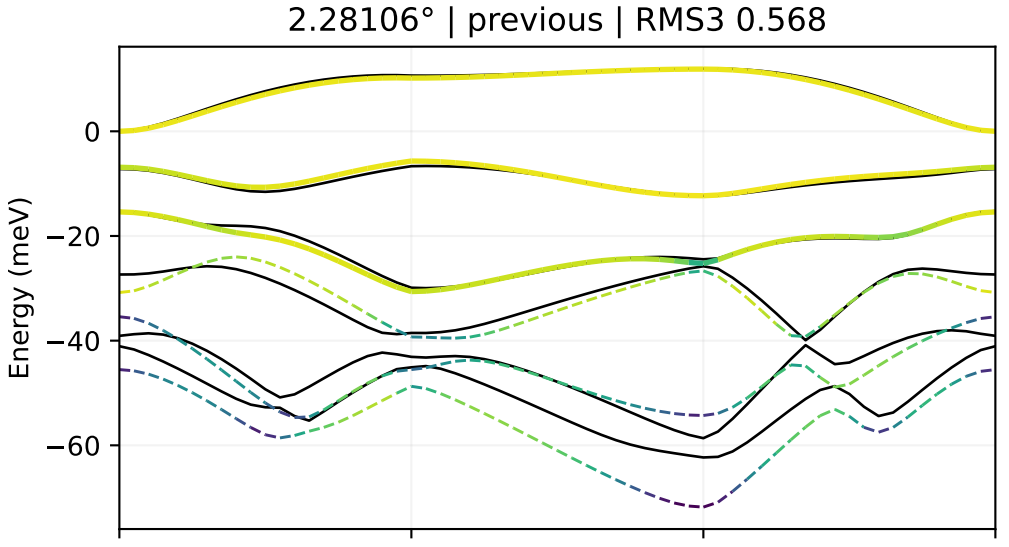


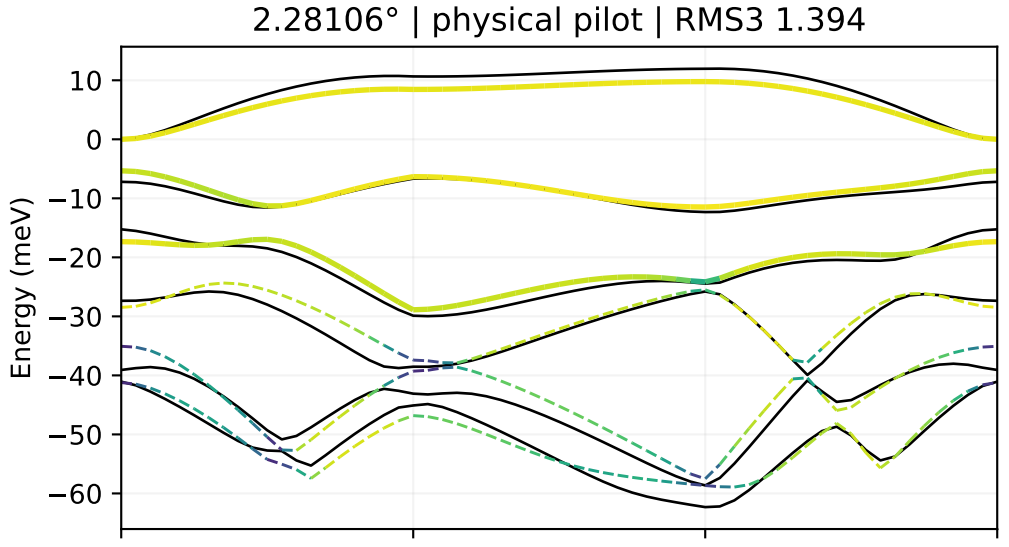

Black: unchanged Heff | colored: model top6 | solid: top3, dashed: 4--6
1.61354° | previous | RMS3 0.289
1.61354° | physical pilot | RMS3 1.029
1.69627° | previous | RMS3 0.283
1.69627° | physical pilot | RMS3 1.022
1.78795° | previous | RMS3 0.334
1.78795° | physical pilot | RMS3 1.019
Energy (meV)
Γ
M
K
Raw same-rank overlap squared

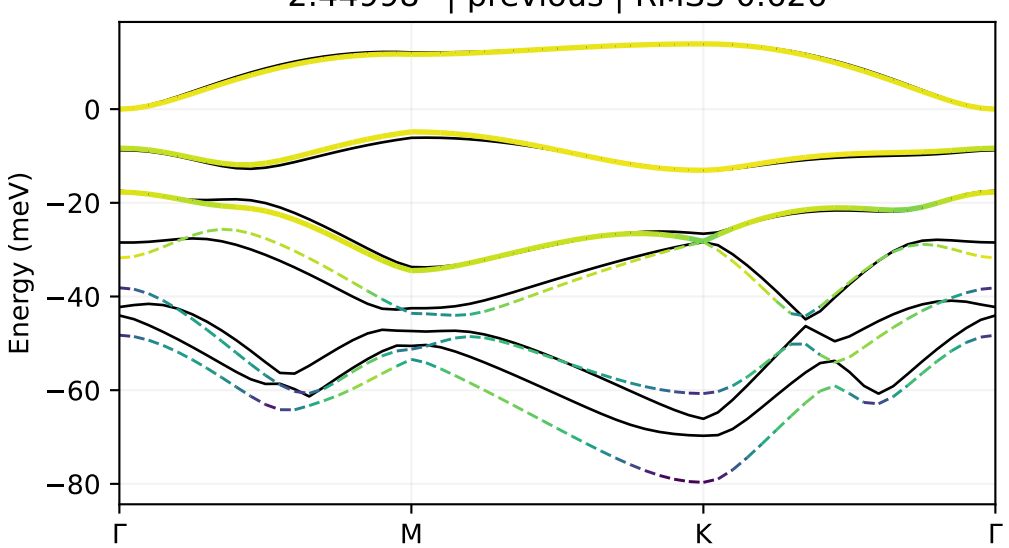
Black: unchanged Heff | colored: model top6 | solid: top3, dashed: 4--6

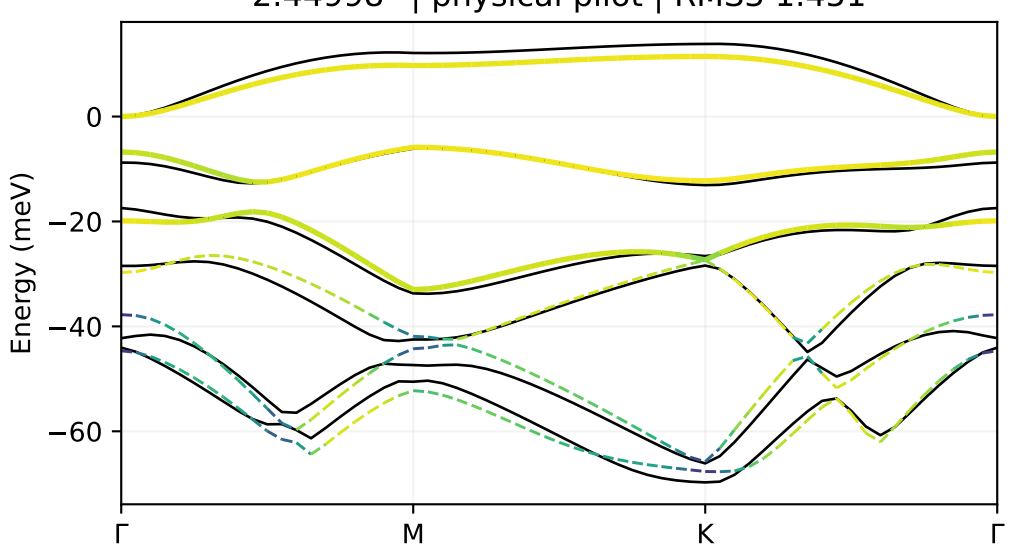


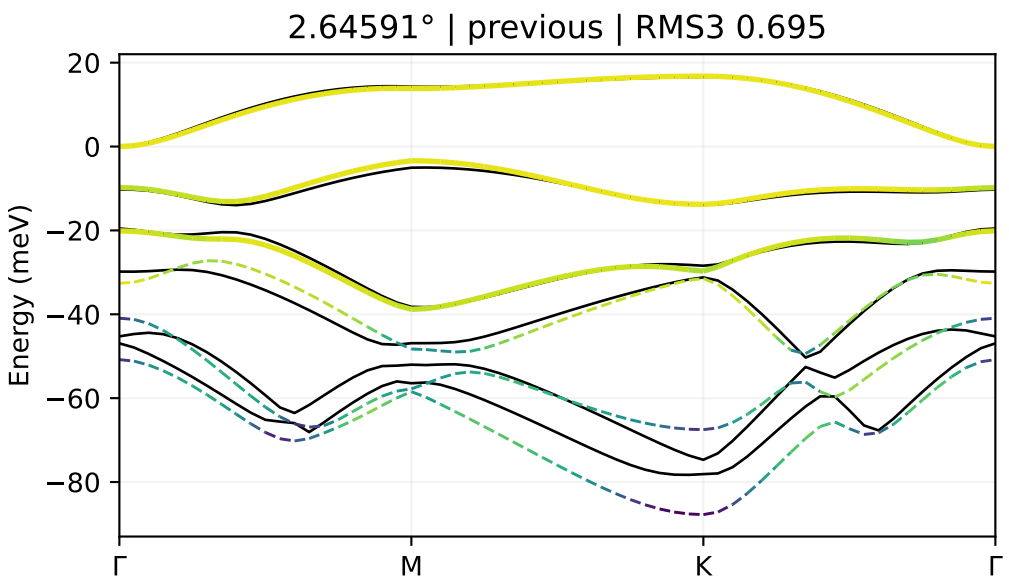


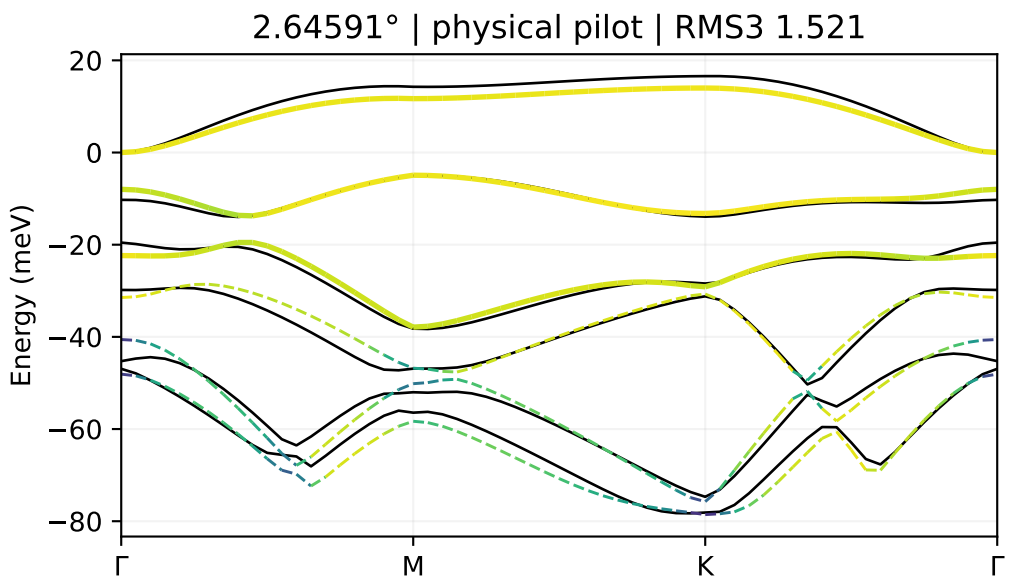


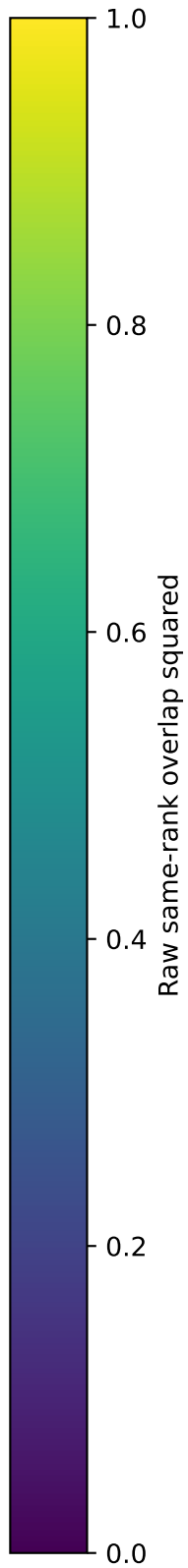




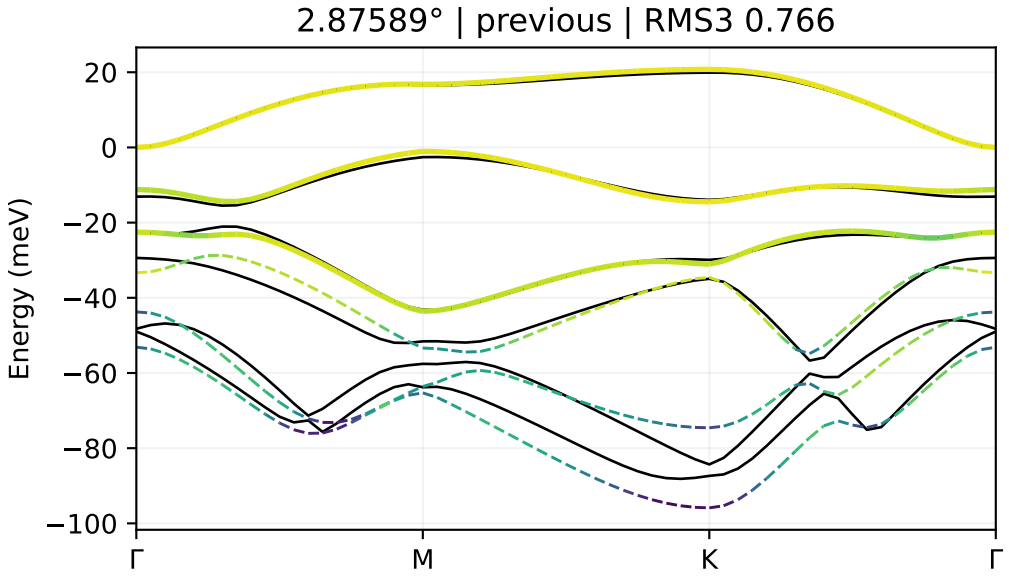


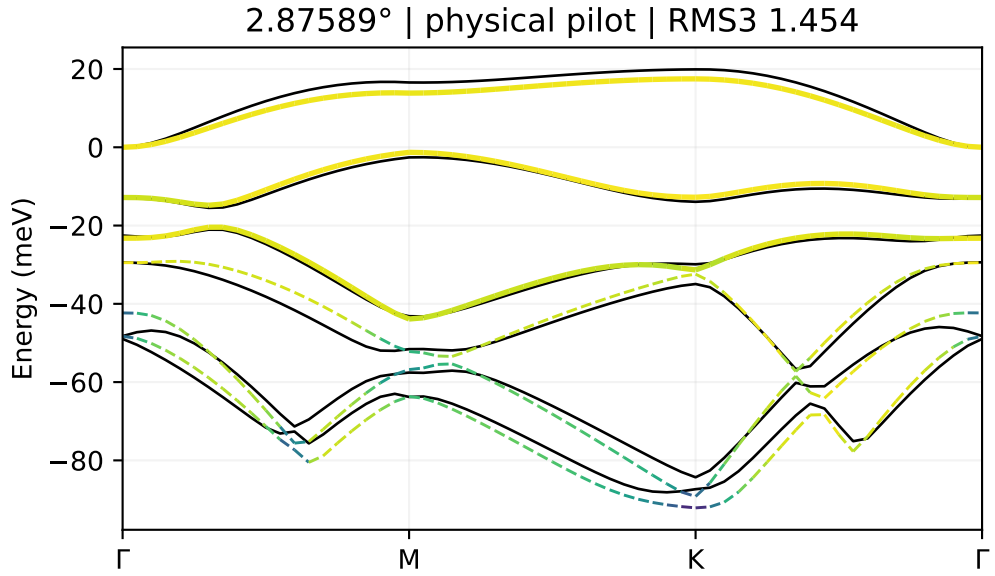


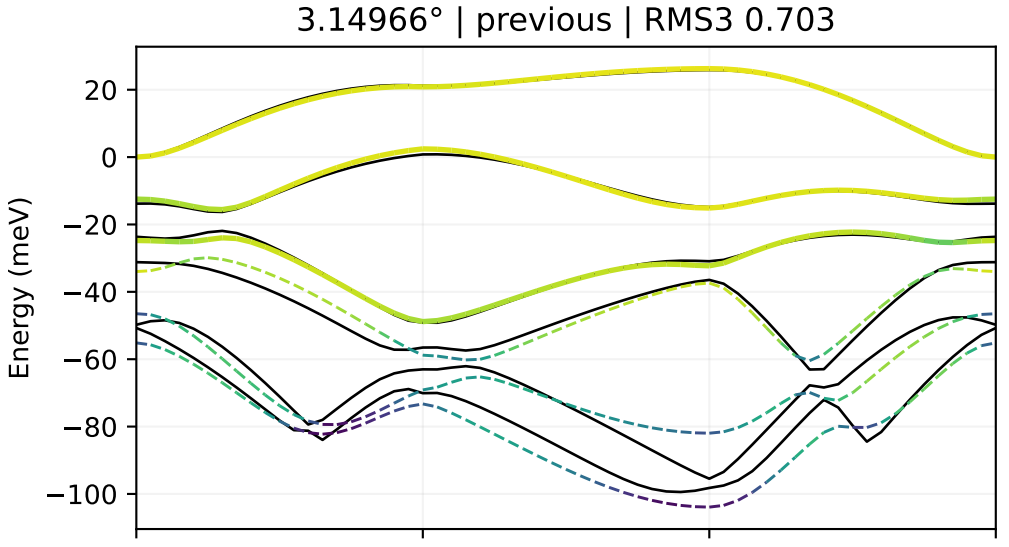


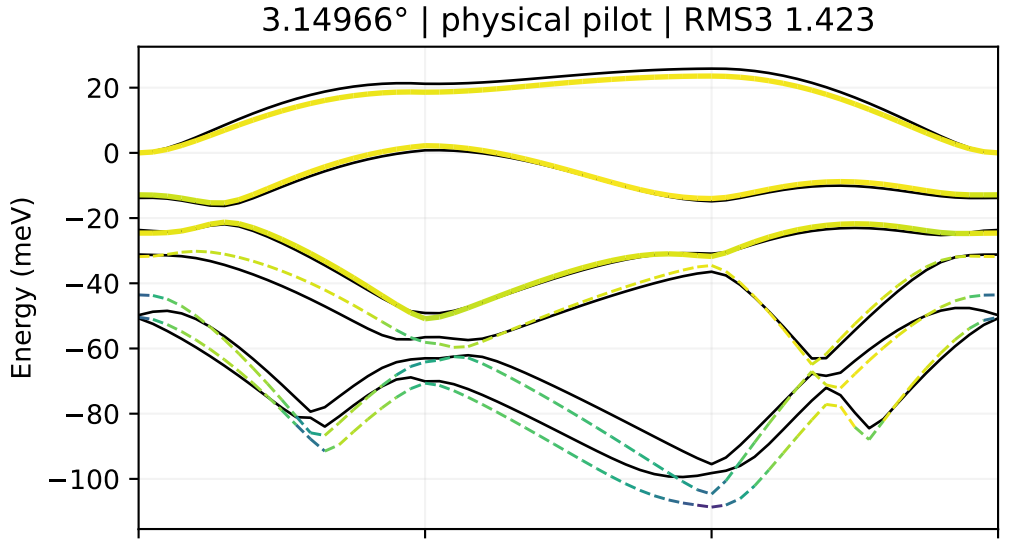


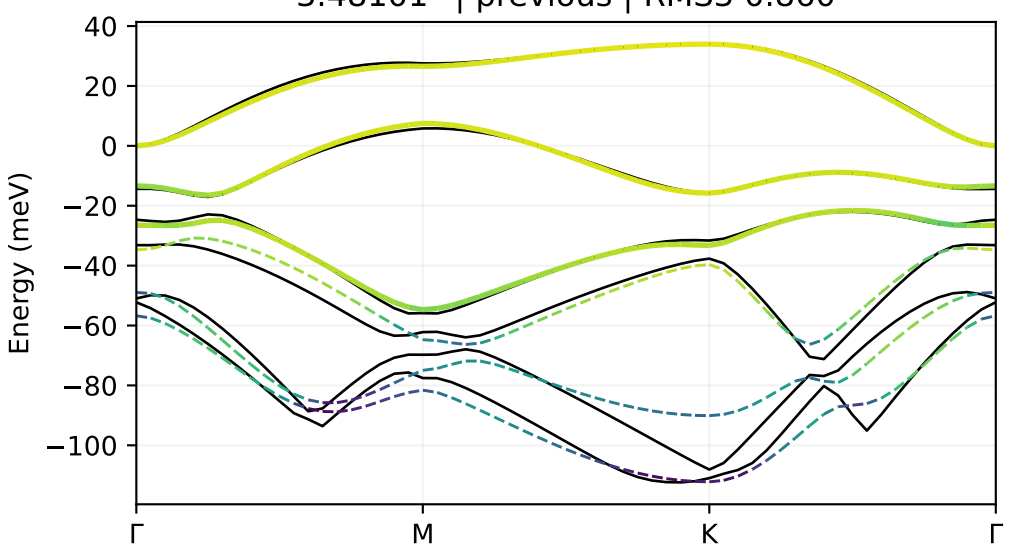
Black: unchanged Heff | colored: model top6 | solid: top3, dashed: 4--6

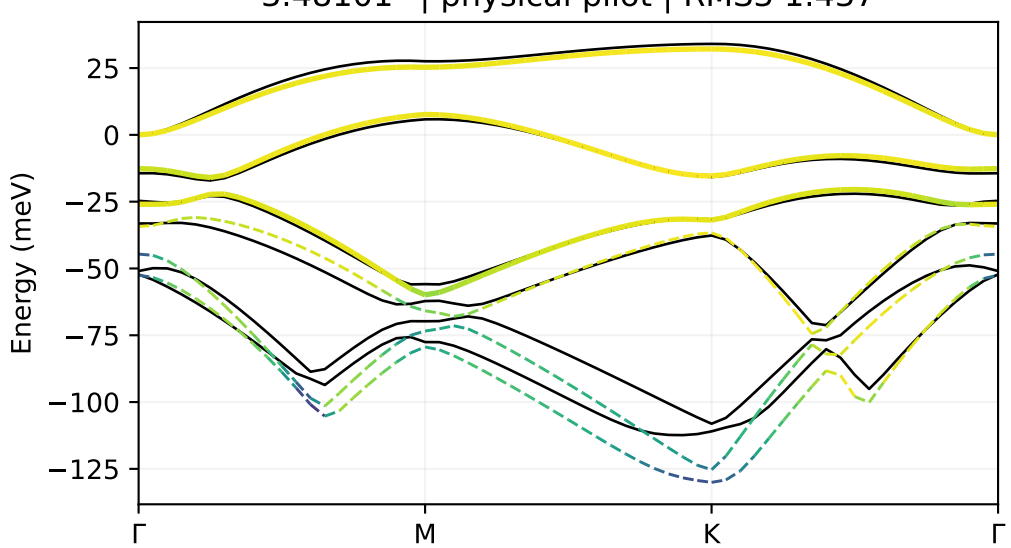


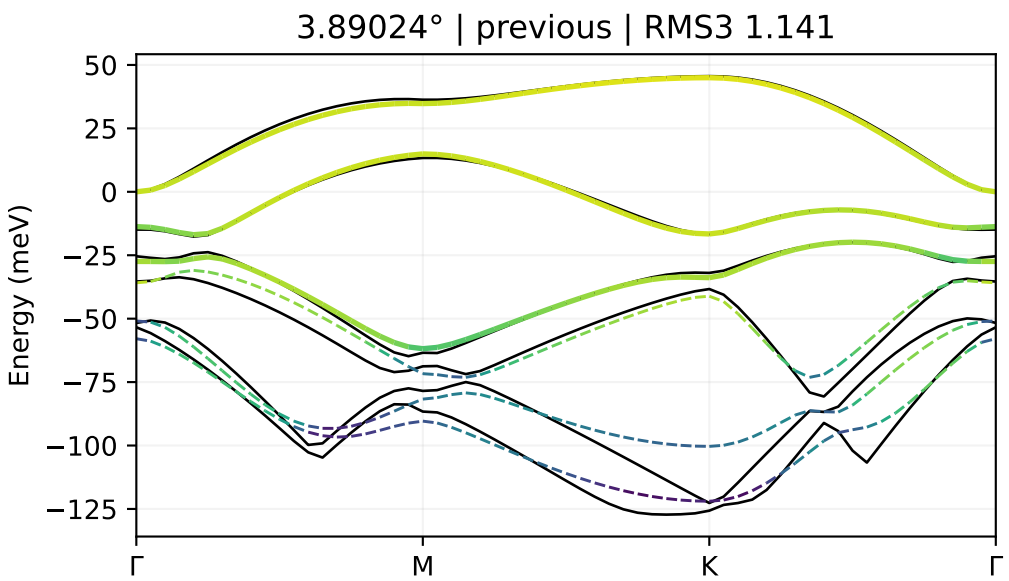


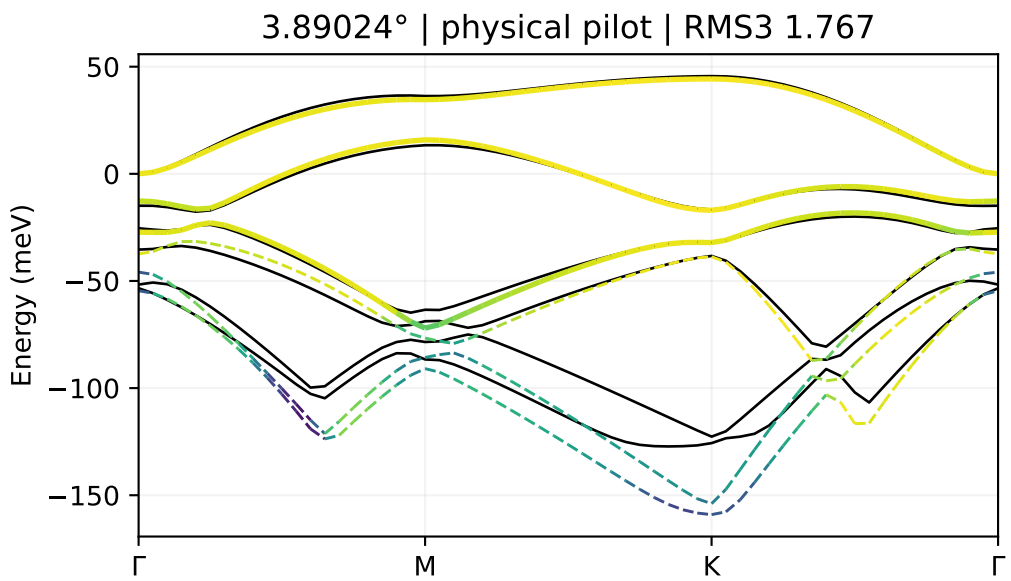


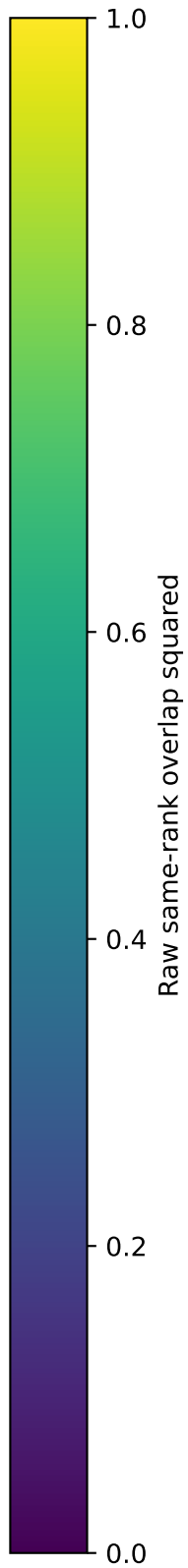


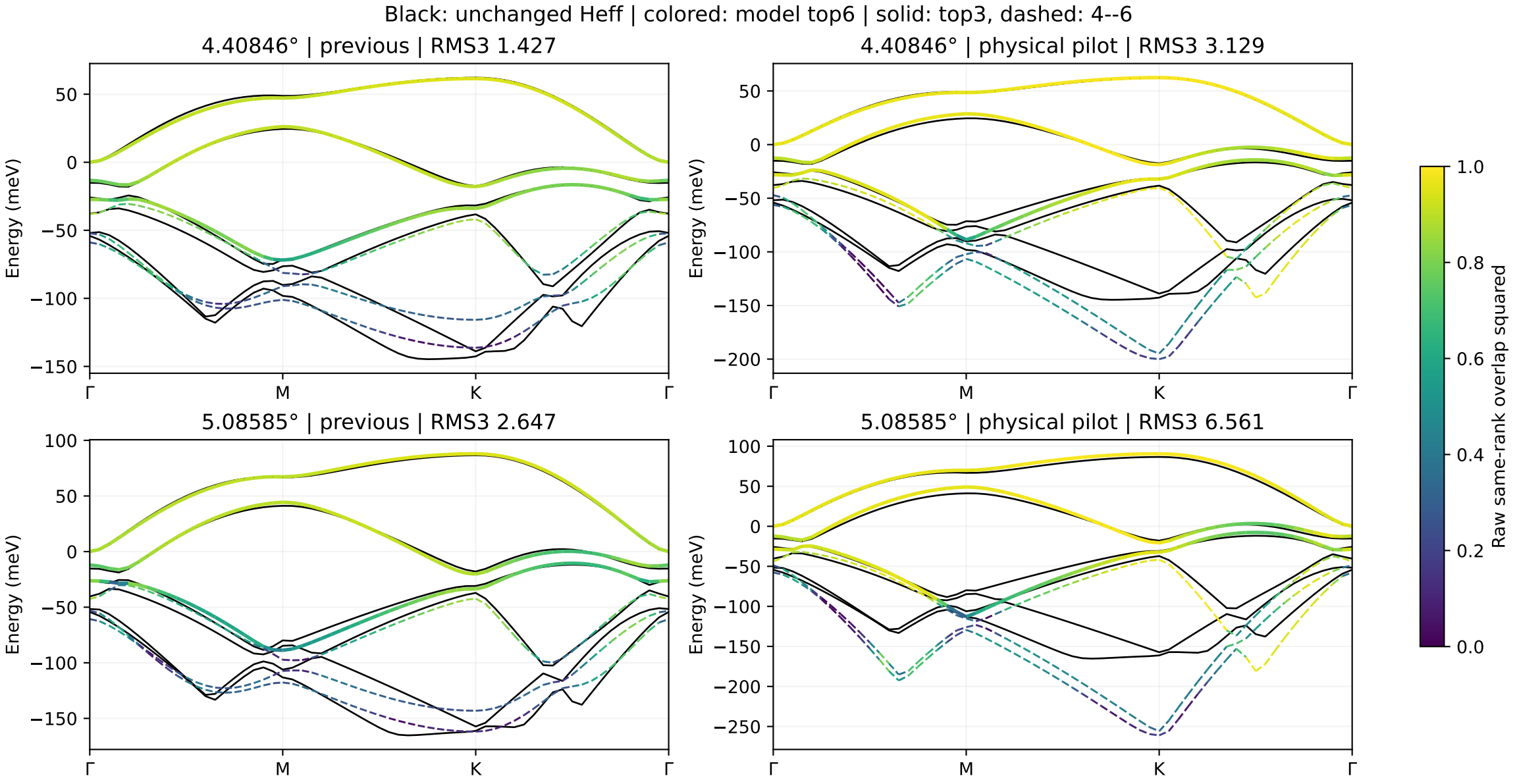


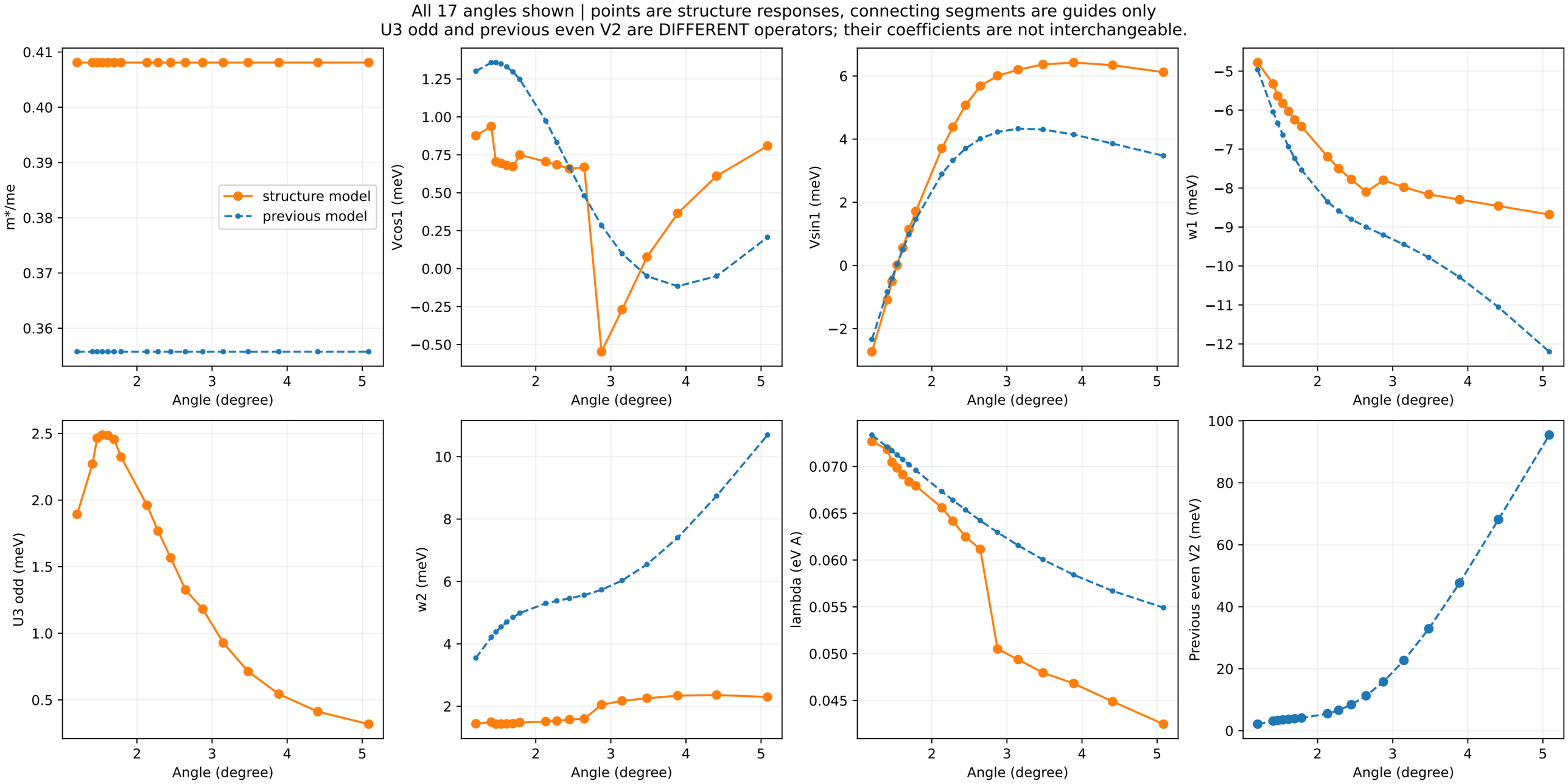




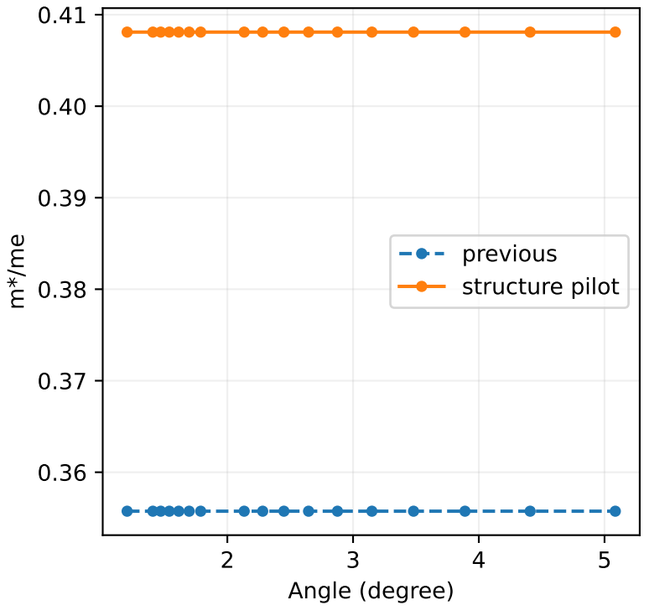


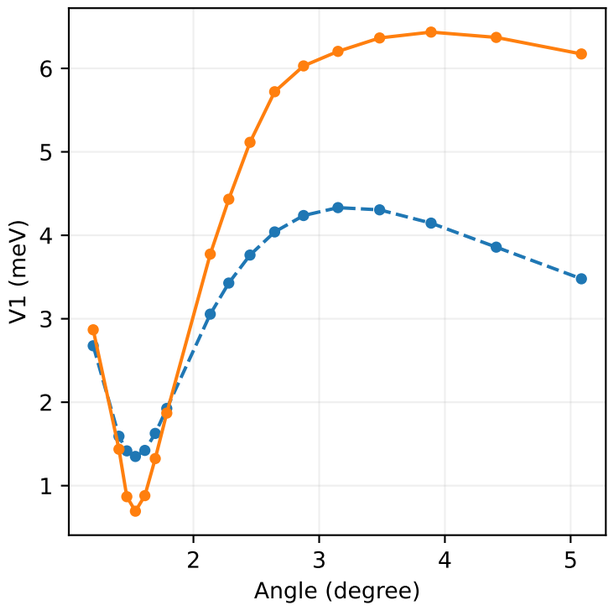


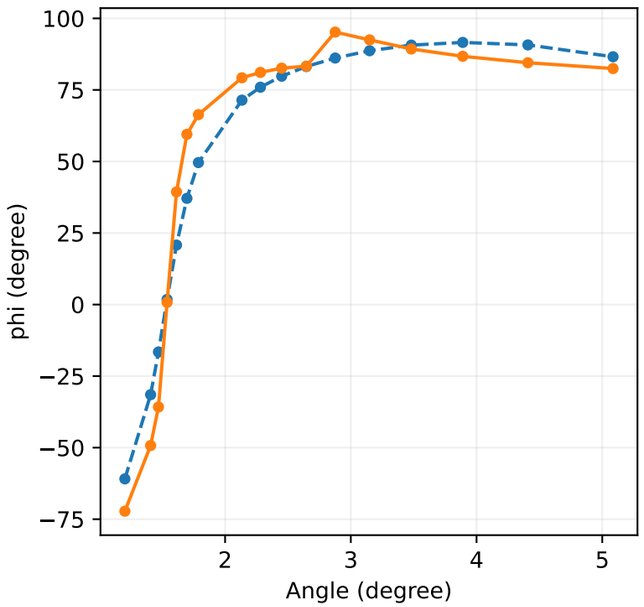

Black: unchanged Heff | colored: model top6 | solid: top3, dashed: 4--6
4.40846° | previous | RMS3 1.427
4.40846° | physical pilot | RMS3 3.129
5.08585° | previous | RMS3 2.647
5.08585° | physical pilot | RMS3 6.561
Energy (meV)
Γ
M
K
Γ
Raw same-rank overlap squared
0.0
0.2
0.4
0.6
0.8
1.0

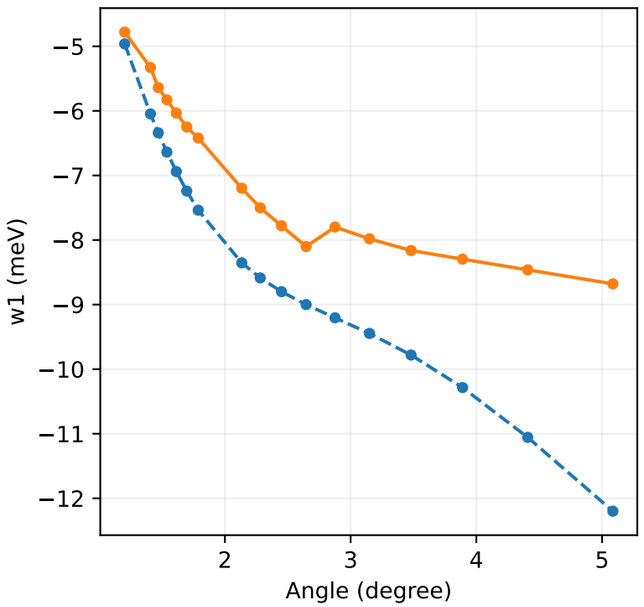

All 17 angles shown | points are structure responses, connecting segments are guides only
U3 odd and previous even V2 are DIFFERENT operators; their coefficients are not interchangeable.
m*/me
structure model
previous model
Vcos1 (meV)
Vsin1 (meV)
w1 (meV)
U3 odd (meV)
w2 (meV)
lambda (eV A)
Previous even V2 (meV)
Angle (degree)

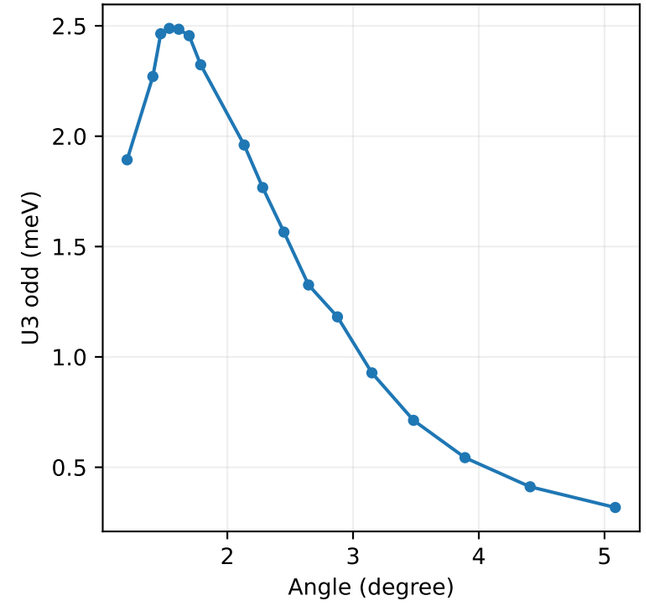
Polar potential convention: V1=sqrt(Vcos1^2+Vsin1^2), phi=atan2(Vsin1,Vcos1)
All calibration points are marked. Lines connect actual structure responses; no extrapolation curve is assumed.

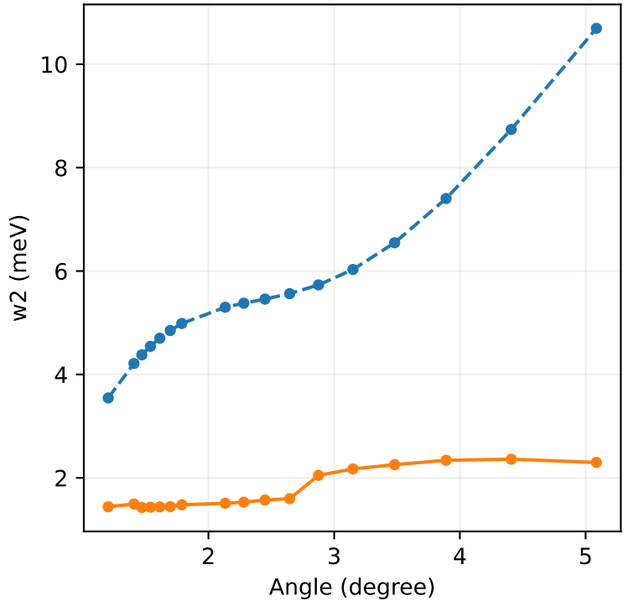


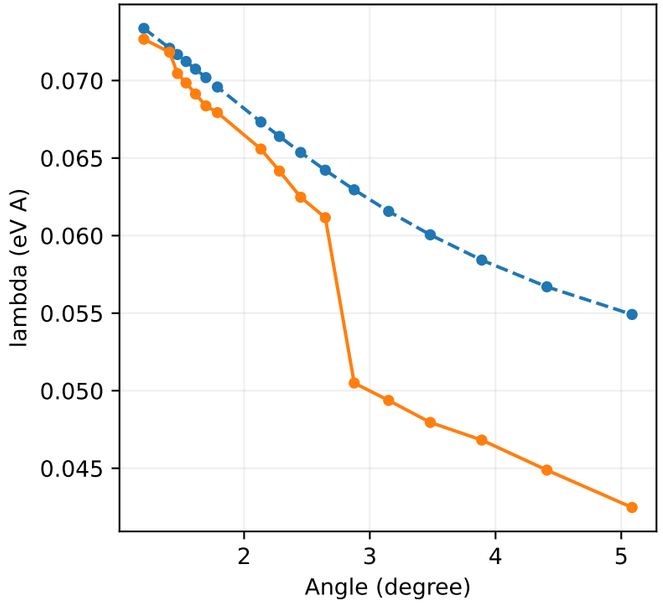


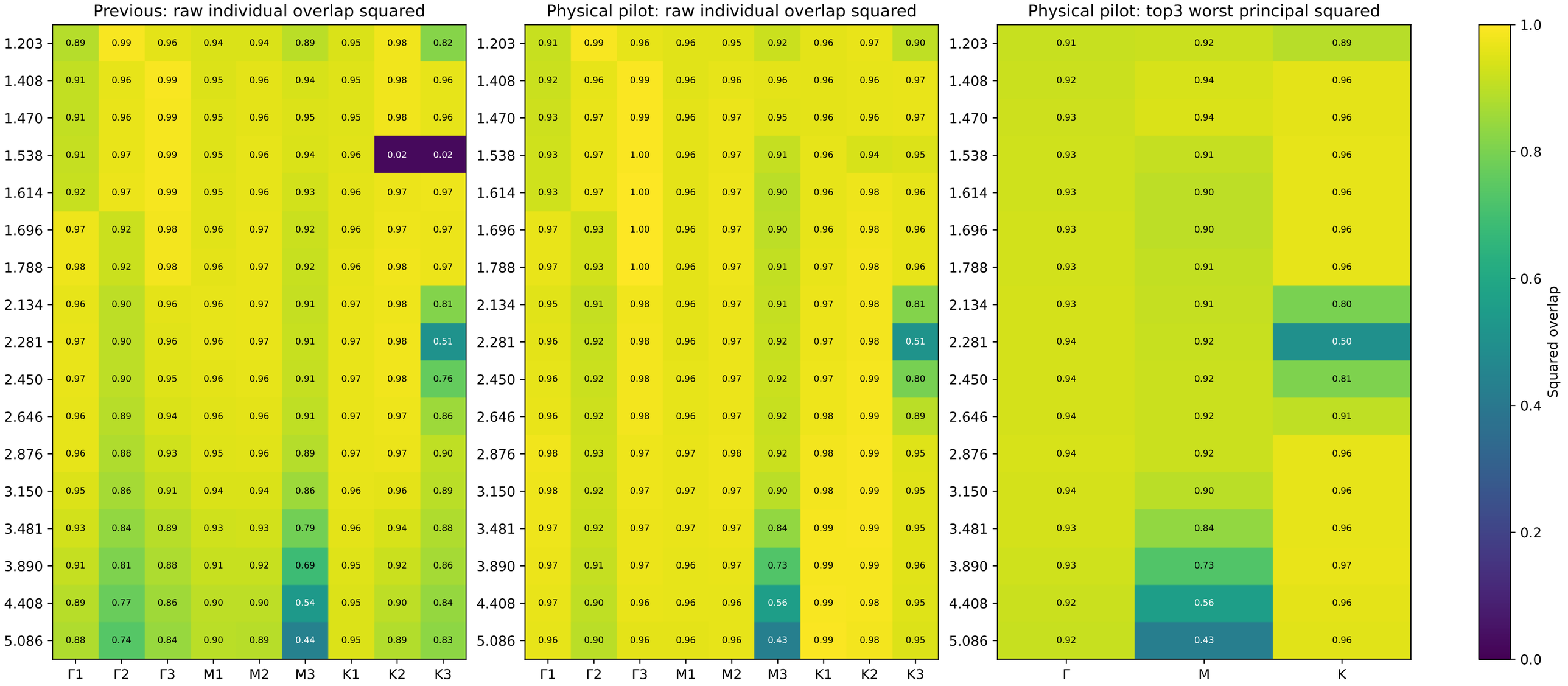


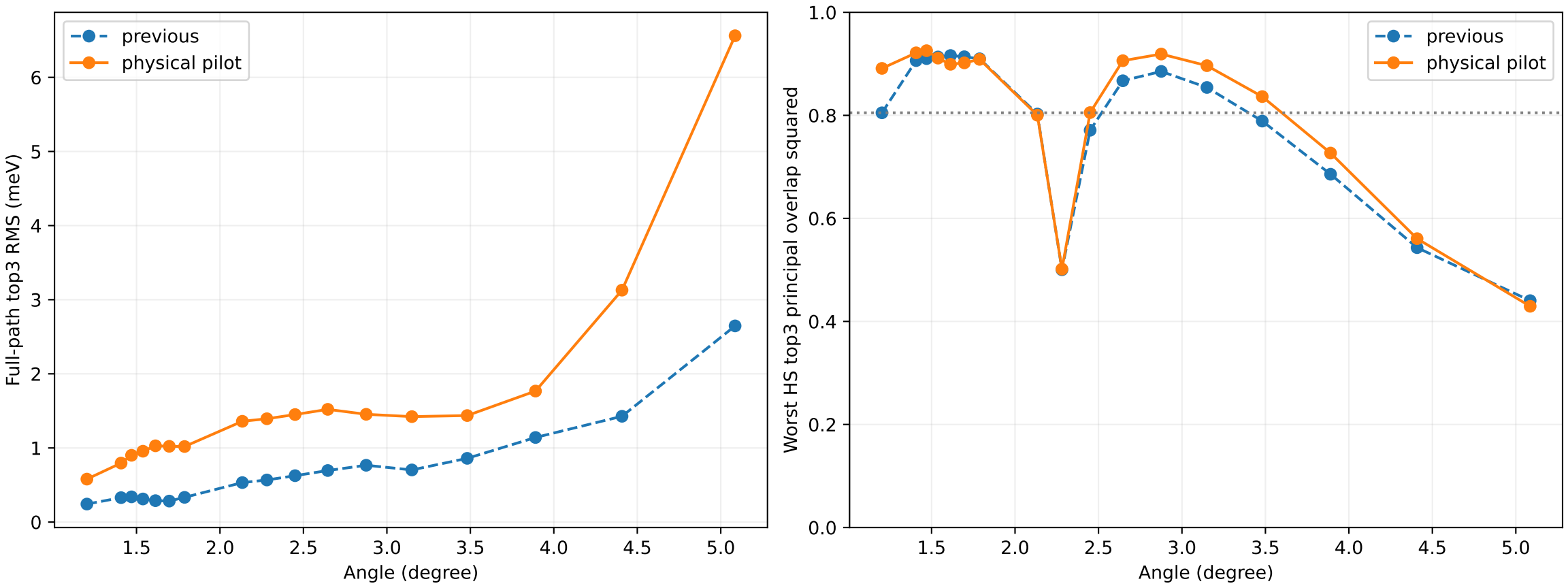


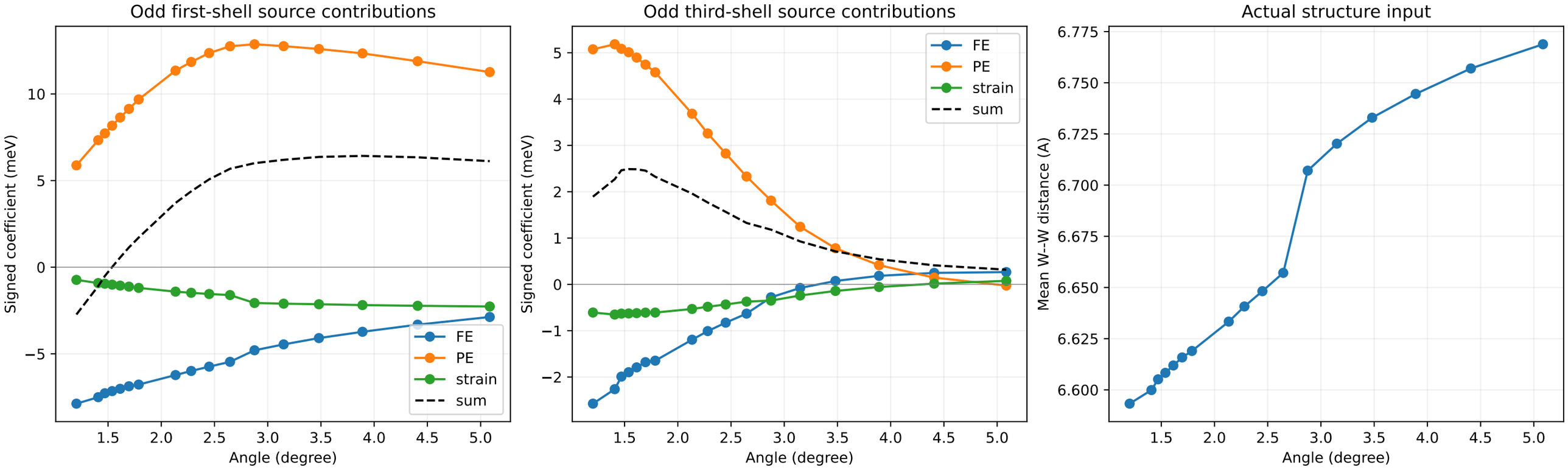


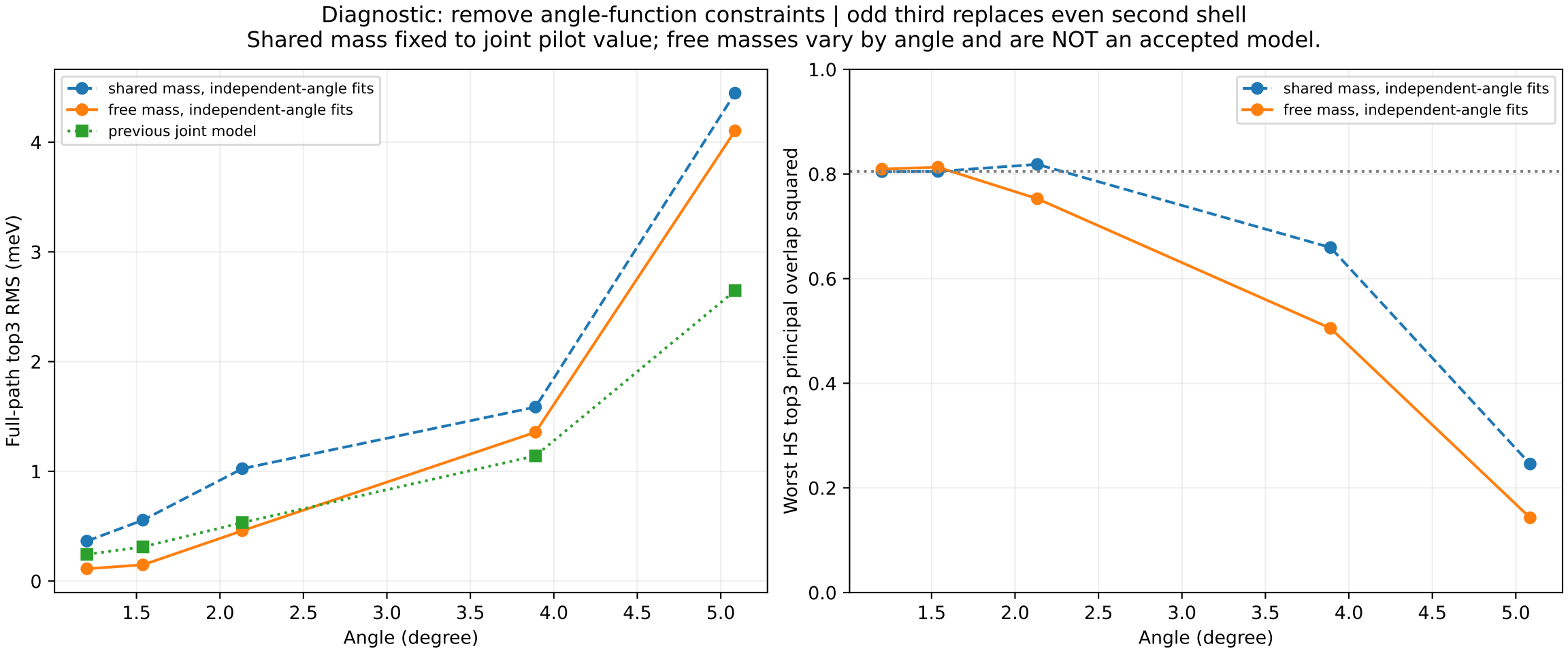





Previous: raw individual overlap squared
Physical pilot: raw individual overlap squared
Physical pilot: top3 worst principal squared
Squared overlap
1.0
0.8
0.6
0.4
0.2
0.0
1.203 0.89 0.99 0.96 0.94 0.94 0.89 0.95 0.98 0.82
1.408 0.91 0.96 0.99 0.95 0.96 0.94 0.95 0.98 0.96
1.470 0.91 0.96 0.99 0.95 0.96 0.95 0.95 0.98 0.96
1.538 0.91 0.97 0.99 0.95 0.96 0.94 0.96 0.02 0.02
1.614 0.92 0.97 0.99 0.95 0.96 0.93 0.96 0.97 0.97
1.696 0.97 0.92 0.98 0.96 0.97 0.92 0.96 0.97 0.97
1.788 0.98 0.92 0.98 0.96 0.97 0.92 0.96 0.98 0.97
2.134 0.96 0.90 0.96 0.96 0.97 0.91 0.97 0.98 0.81
2.281 0.97 0.90 0.96 0.96 0.97 0.91 0.97 0.98 0.51
2.450 0.97 0.90 0.95 0.96 0.96 0.91 0.97 0.98 0.76
2.646 0.96 0.89 0.94 0.96 0.96 0.91 0.97 0.97 0.86
2.876 0.96 0.88 0.93 0.95 0.96 0.89 0.97 0.97 0.90
3.150 0.95 0.86 0.91 0.94 0.94 0.86 0.96 0.96 0.89
3.481 0.93 0.84 0.89 0.93 0.93 0.79 0.96 0.94 0.88
3.890 0.91 0.81 0.88 0.91 0.92 0.69 0.95 0.92 0.86
4.408 0.89 0.77 0.86 0.90 0.90 0.54 0.95 0.90 0.84
5.086 0.88 0.74 0.84 0.90 0.89 0.44 0.95 0.89 0.83
Γ1 Γ2 Γ3 M1 M2 M3 K1 K2 K3
1.203 0.91 0.99 0.96 0.96 0.95 0.92 0.96 0.97 0.90
1.408 0.92 0.96 0.99 0.96 0.96 0.96 0.96 0.96 0.97
1.470 0.93 0.97 0.99 0.96 0.97 0.95 0.96 0.96 0.97
1.538 0.93 0.97 1.00 0.96 0.97 0.91 0.96 0.94 0.95
1.614 0.93 0.97 1.00 0.96 0.97 0.90 0.96 0.98 0.96
1.696 0.97 0.93 1.00 0.96 0.97 0.90 0.96 0.98 0.96
1.788 0.97 0.93 1.00 0.96 0.97 0.91 0.97 0.98 0.96
2.134 0.95 0.91 0.98 0.96 0.97 0.91 0.97 0.98 0.81
2.281 0.96 0.92 0.98 0.96 0.97 0.92 0.97 0.98 0.51
2.450 0.96 0.92 0.98 0.96 0.97 0.92 0.97 0.99 0.80
2.646 0.96 0.92 0.98 0.96 0.97 0.92 0.98 0.99 0.89
2.876 0.98 0.93 0.97 0.97 0.98 0.92 0.98 0.99 0.95
3.150 0.98 0.92 0.97 0.97 0.97 0.90 0.98 0.99 0.95
3.481 0.97 0.92 0.97 0.97 0.97 0.84 0.99 0.99 0.95
3.890 0.97 0.91 0.97 0.96 0.97 0.73 0.99 0.99 0.96
4.408 0.97 0.90 0.96 0.96 0.96 0.56 0.99 0.98 0.95
5.086 0.96 0.90 0.96 0.96 0.96 0.43 0.99 0.98 0.95
Γ1 Γ2 Γ3 M1 M2 M3 K1 K2 K3
1.203 0.91 0.92 0.89
1.408 0.92 0.94 0.96
1.470 0.93 0.94 0.96
1.538 0.93 0.91 0.96
1.614 0.93 0.90 0.96
1.696 0.93 0.90 0.96
1.788 0.93 0.91 0.96
2.134 0.93 0.91 0.80
2.281 0.94 0.92 0.50
2.450 0.94 0.92 0.81
2.646 0.94 0.92 0.91
2.876 0.94 0.92 0.96
3.150 0.94 0.90 0.96
3.481 0.93 0.84 0.96
3.890 0.93 0.73 0.97
4.408 0.92 0.56 0.96
5.086 0.92 0.43 0.96
Γ M K

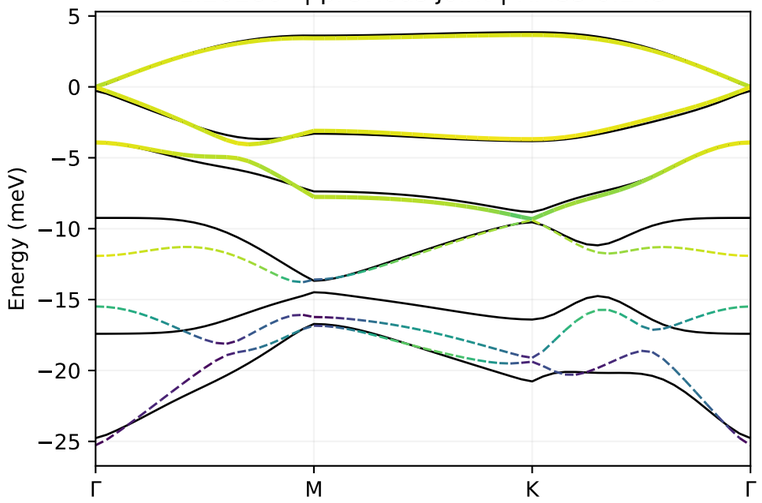

previous
physical pilot
Full-path top3 RMS (meV)
Angle (degree)
0
1
2
3
4
5
6
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
previous
physical pilot
Worst HS top3 principal overlap squared
Angle (degree)
0.0
0.2
0.4
0.6
0.8
1.0


Odd first-shell source contributions
Signed coefficient (meV)
Angle (degree)
FE
PE
strain
sum
Odd third-shell source contributions
Signed coefficient (meV)
Angle (degree)
FE
PE
strain
sum
Actual structure input
Mean W–W distance (A)
Angle (degree)

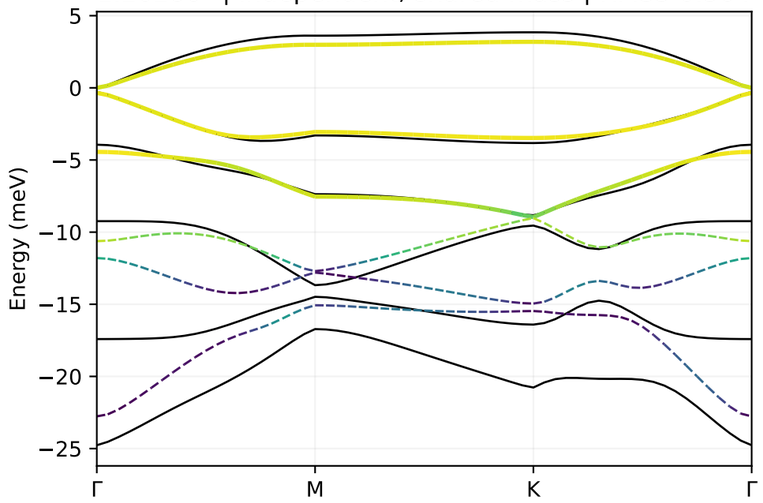

Diagnostic: remove angle-function constraints | odd third replaces even second shell
Shared mass fixed to joint pilot value; free masses vary by angle and are NOT an accepted model.
shared mass, independent-angle fits
free mass, independent-angle fits
previous joint model
Full-path top3 RMS (meV)
Angle (degree)
Worst HS top3 principal overlap squared
Angle (degree)

Operator-budget diagnostic, NOT proposed independent-angle parameters
No Heff symmetrization; all six bands shown; enlarged-basis full-path audit.

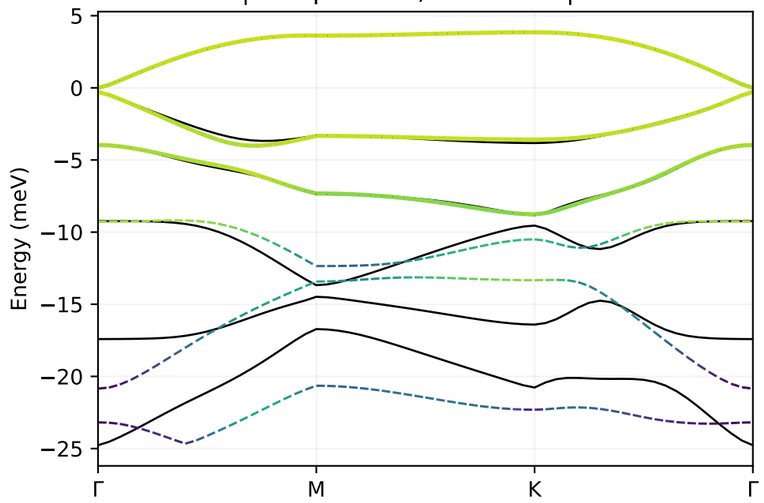
1.20286° | previous joint | RMS3 0.243



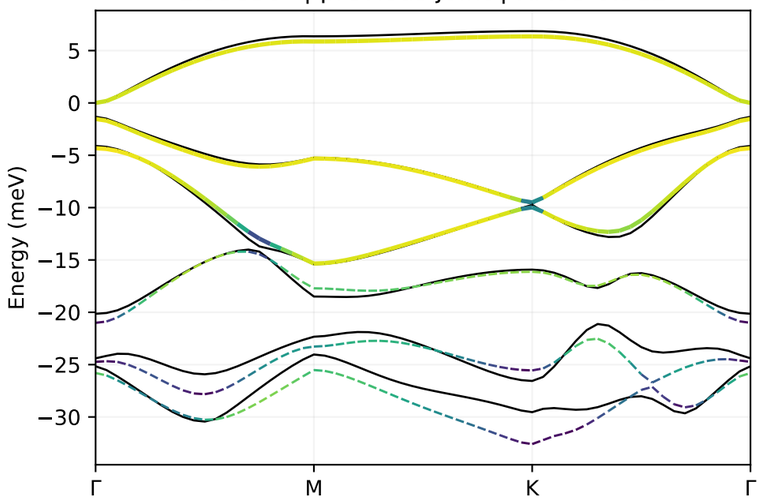
1.20286° | independent, shared mass | RMS3 0.364



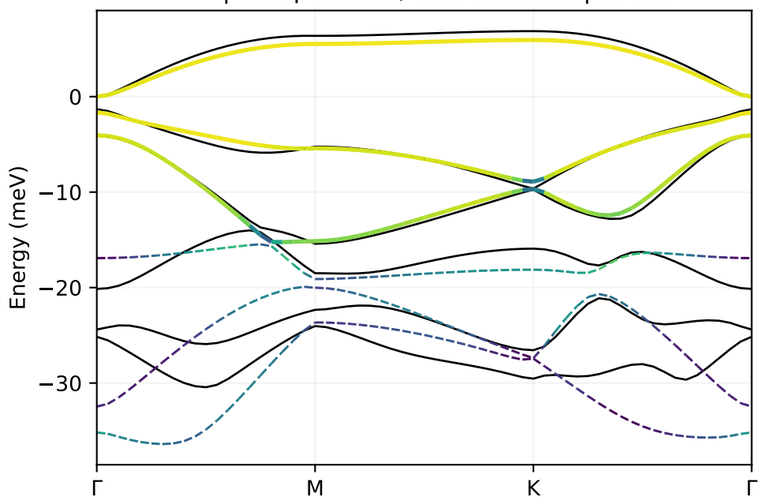
1.20286° | independent, free mass | RMS3 0.112



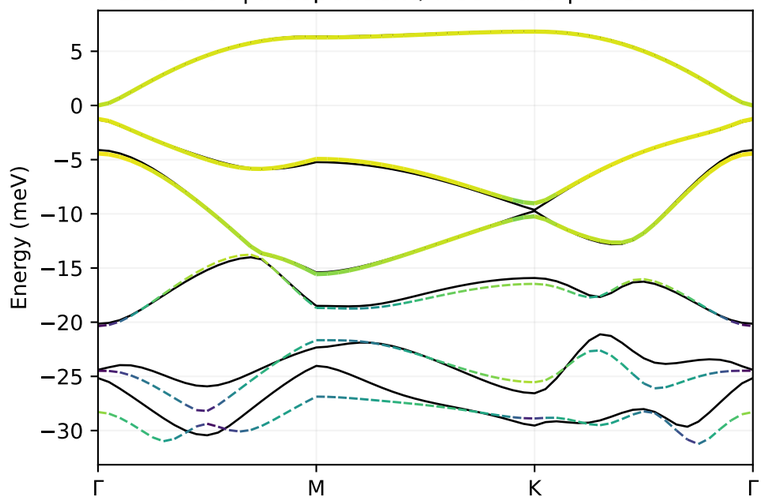
1.53850° | previous joint | RMS3 0.312



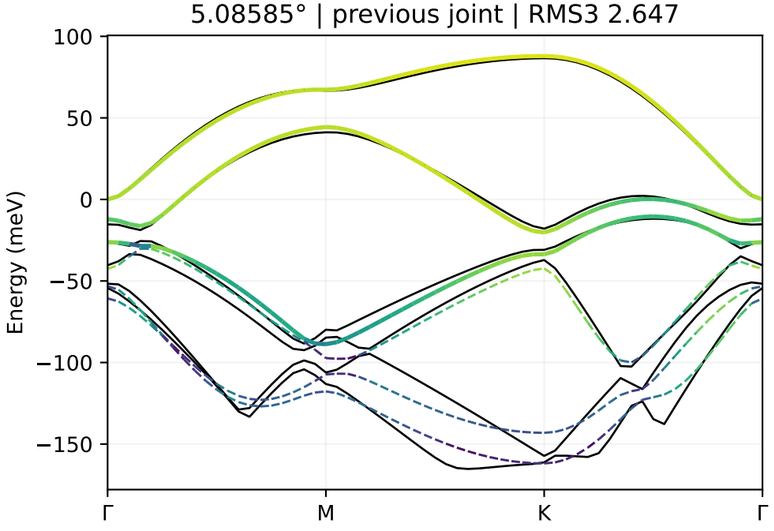
1.53850° | independent, shared mass | RMS3 0.555



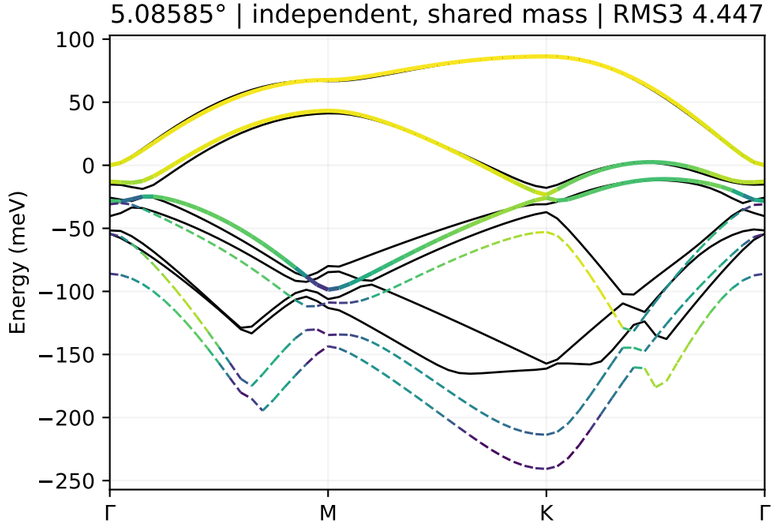
1.53850° | independent, free mass | RMS3 0.148



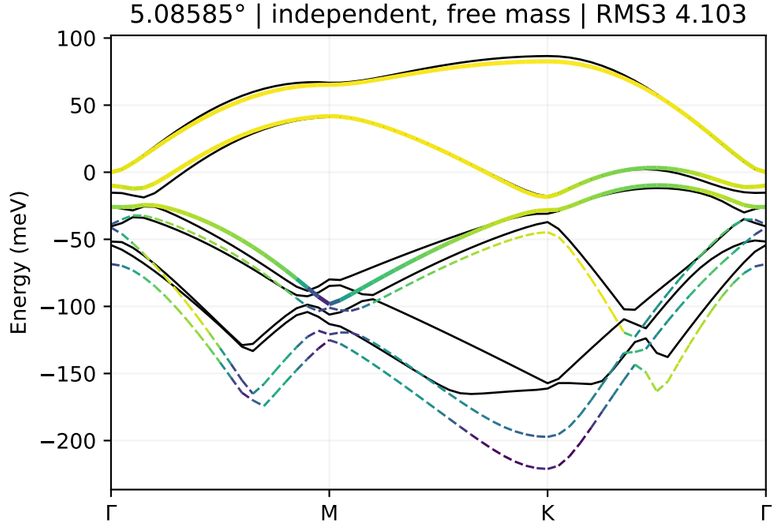
5.08585° | previous joint | RMS3 2.647

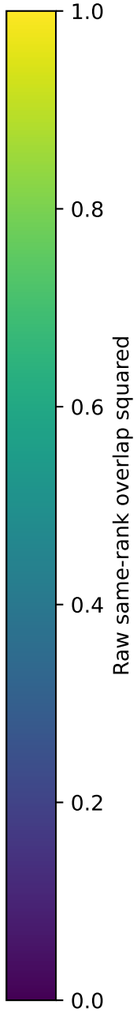


5.08585° | independent, shared mass | RMS3 4.447



5.08585° | independent, free mass | RMS3 4.103





Controlled operator reallocation, always SEVEN fitted coefficients per angle
Ablation only: independent angles and masses. No new global or extrapolation model is claimed.

